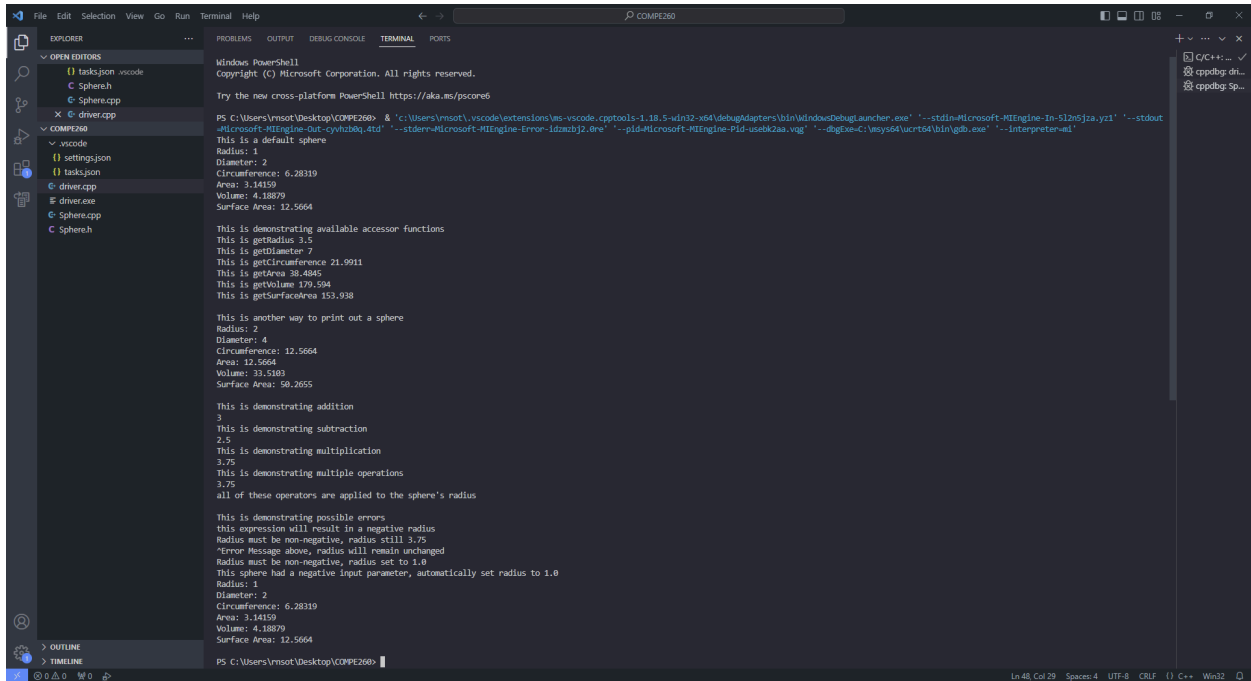


Programming Assignment 1: Classes

[Screenshot of Runtime Output of Code]



```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\Users\vnsoot\Desktop\COMP260> & 'C:\Users\vnsoot\.vscode\extensions\ms-vscode.cpptools-1.18.5-win32-x64\debugadapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-S12n5jpa.yri1' '--stdout=Microsoft-MIEngine-Out-cyhzb0q.dtd' '--stderr=Microsoft-MIEngine-Error-ldzchj2.0ne' '--pid=Microsoft-MIEngine-Pid-usek2aa.vvg' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=ml'

This is a default sphere
Radius: 1
Diameter: 2
Circumference: 6.28319
Area: 3.14159
Volume: 4.18879
Surface Area: 12.5664

This is demonstrating available accessor functions
This is getRadius 3.5
This is getDiameter 7
This is getCircumference 21.9911
This is getArea 38.4845
This is getVolume 179.594
This is getSurfaceArea 153.938

This is another way to print out a sphere
Radius: 2
Diameter: 4
Circumference: 12.5664
Area: 12.5664
Volume: 13.5103
Surface Area: 50.2655

This is demonstrating addition
3
This is demonstrating subtraction
2.5
This is demonstrating multiplication
3.75
This is demonstrating multiple operations
3.75
all of these operators are applied to the sphere's radius

This is demonstrating possible errors
this expression will result in a negative radius
Radius must be non-negative, radius still 3.75
*Error Message above, radius will remain unchanged
Radius must be non-negative, radius set to 1.0
This sphere had a negative input parameter, automatically set radius to 1.0
Radius: 1
Diameter: 2
Circumference: 6.28319
Area: 3.14159
Volume: 4.18879
Surface Area: 12.5664

PS C:\Users\vnsoot\Desktop\COMP260>
```

DISCUSSION

Comments:

My program is doing everything that it is designed to do, the driver program tests every member and non-member function of the class, and is itself thoroughly commented on.

Corner Cases:

I have tested all of my functions and overloaded operators to work with my sphere class. Here is a list of things that I didn't test however:

- Extremely *large* values for radius
- Extremely *small* values for radius

I have attempted to prevent the user from using negative values for sphere radius, adding in checks for anytime a sphere's radius is altered in anyway, i.e. making the radius a negative value through overloaded operators or the explicit value constructor will either prevent the radius from being changed or set the radius to the default value respectively.
